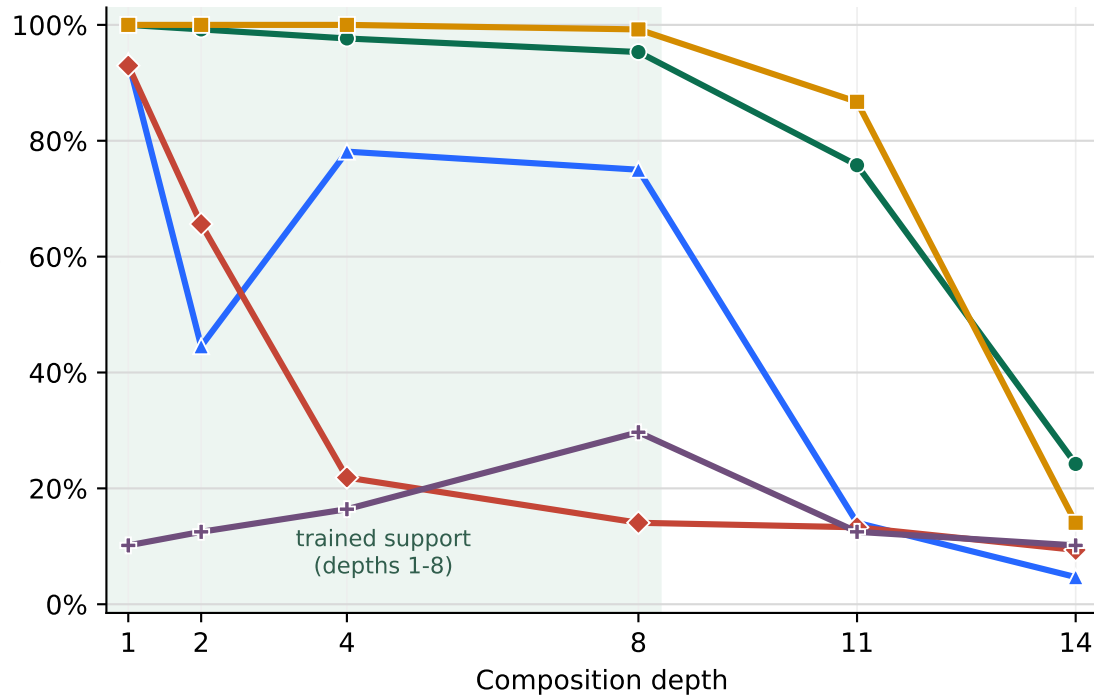


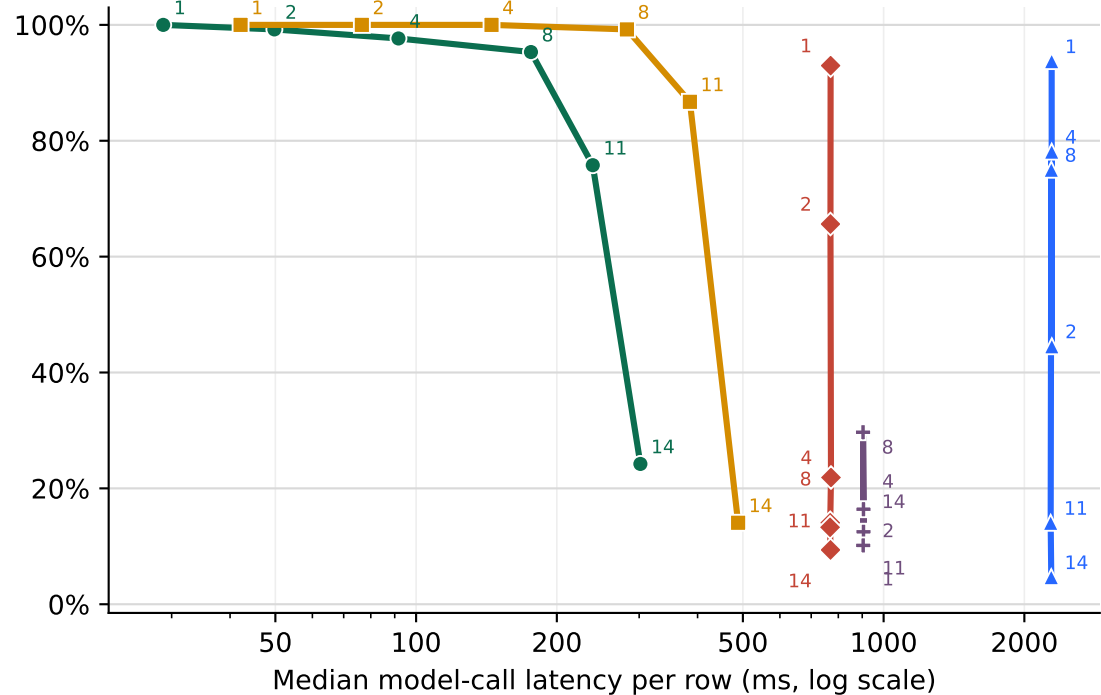
Accuracy, recurrent depth, and wall-clock cost on the frozen Phase A family

Same 128 rows per depth; complete synchronized model-call latency on one A100, batch size 1

A. Accuracy by composition depth



B. Accuracy by registered-path latency



● Full-block recurrent ■ Adapter recurrent ▲ Dense scratchpad 0.5B ◆ Dense direct 0.5B + Dense direct 1.5B